

Scale-SETI Project Roadmap and Scope Control

Publication, Engineering, Review, and Experimental Plan

Version 1.1

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If a new item does not close a defined publication, reproducibility, review, or experimental gap, it does not belong in the current artifact.

1. Purpose

This roadmap prevents scope drift and defines what each Scale-SETI artifact is supposed to accomplish. The project will no longer add material simply because additional detail is possible. Each document has a defined audience, stopping condition, and next action.

2. Project Architecture

Product A - Core Concept Paper

Purpose: Present the intellectual contribution, scientific motivation, falsifiable research question, evidence standards, relationship to prior work, and high-level experimental roadmap.

Primary audience: Physicists, SETI/technosignature researchers, information theorists, experimentalists, reviewers, and potential collaborators.

Baseline: Revision 5 technical-review edition.

What belongs here:

- central Scale-SETI hypothesis and search-space framing
- established science versus speculative extensions
- minimal scale-coordinate framework
- relationship to prior work
- falsifiability and evidence ladder
- high-level Phase-1 experimental design
- major objections and limitations
- publication-quality references

- up-scale SETI as a conceptual/distributed-analysis branch distinct from the Phase-1 microscopic bench
- bidirectional macro/micro model showing that our regime may be intermediate rather than privileged
- wall analogy as the accessible conceptual explanation of a shared boundary interface
- expanded SETI search-space framework: same-scale, down-scale, and up-scale search

What does not belong here:

- connector pinouts
- full wiring schedules
- detailed procurement BOMs
- calibration worksheets
- software repository internals
- fabrication drawings
- step-by-step bench assembly

Target size/scope: 25-35 pages including references and essential figures.

Definition of done:

1. hypothesis clear and internally consistent
2. established science separated from speculation
3. novelty and prior work accurately represented
4. falsifiable proposition explicit
5. evidence and failure criteria defined
6. major objections addressed
7. external reviewers given opportunity to critique
8. remaining changes limited to scientific accuracy, clarity, or reviewer response

Product B - Phase-1 Technical Design Package

Purpose: Allow an independent instrumentation/microfluidics laboratory to reproduce the first controlled experiment without guessing important design decisions.

Primary audience: Instrumentation engineers, microfluidics researchers, electronics engineers, experimental physicists, software/data engineers, and replication labs.

Baseline: Phase-1 Design Package v0.3.

What belongs here:

- complete architecture
- electrical and fluidic schematics
- wiring and I/O maps
- mechanical/fabrication drawings
- BOM and alternates
- host/network isolation design
- software interfaces and schemas
- calibration procedures
- control matrix
- acceptance criteria
- bring-up procedure
- troubleshooting and replication instructions
- release manifests and hashes

Target size/scope: As long as needed for complete reproducibility.

Definition of done:

1. critical components identified by part number or performance specification
2. every intentional signal path documented
3. relevant leakage paths mitigated or measured
4. all electrical/fluidic interfaces documented
5. calibration and acceptance procedures executable
6. data schemas and interfaces validated
7. another qualified lab can build from the package
8. unresolved decisions non-critical or explicitly documented

Product C - Experimental Results Paper

Purpose: Report actual experimental results, including null results, after commissioning and preregistration.

Primary audience: Scientific reviewers and the broader research community.

Baseline: Created only after real data exist.

What belongs here:

- final apparatus used
- deviations from Design Package 1.0
- preregistered hypotheses and analysis

- calibration and acceptance results
- control results
- blinded trial results
- statistics and uncertainty
- anomaly investigation
- raw-data availability
- independent replication status

Target size/scope: Determined by results and target venue.

Definition of done:

1. real experimental data exist
2. analysis follows preregistration
3. results and controls are complete
4. data and code are archived

3. Current Project Status

Milestone	Status	Notes
Original idea	Complete	Search for intelligence across physical scale
Conceptual hypothesis	Complete	Hierarchical / nonlinear scale framing
Scientific framing	Complete	Established science separated from speculation
Falsifiability framework	Complete	Challenge-response and control hierarchy defined
Formal technical review	Complete	Revision 5
Phase-1 reference architecture	Complete	Known-physics validation bench
Engineering control documents	Complete	Through v0.3
Stage-1 implementation choice	Complete	Ion concentration / microfluidic reference channel
Dry-run software/data pipeline	Complete	Synthetic campaign demonstrated
Fabrication/build release	NEXT	Close remaining reproducibility gaps
External expert review	Pending	NEXT publication gate
Concept Paper Candidate 1.0	Complete	Expanded same-scale/down-scale/up-scale framework; ready for external review

Public preprint / DOI	Pending	Zenodo / OSF first
Physical prototype	Pending	After design review
Commissioning	Pending	Acceptance gates before search experiment
Search experiment	Pending	Only after validated baseline
Results paper	Pending	Based on real data

Scope boundary added in v1.1: the up-scale SETI concept belongs to Product A. It does not reopen the Phase-1 microscopic engineering scope. Any future distributed up-scale sensor/probe architecture is deferred until after Concept Paper 1.0 review and publication planning.

4. Remaining Work Before External Review

Complete only the engineering items necessary for a competent external reviewer to evaluate reproducibility and failure modes.

- dimensioned receiver-cell / flow-cell drawing
- complete electrical schematic with component values
- grounding, shielding, and isolation schematic
- fluidic assembly drawing
- connector and pin schedule
- final procurement BOM with quantities, alternates, and performance requirements
- physical cable-routing and zone-boundary drawing
- bench assembly procedure
- bench bring-up sequence
- troubleshooting matrix
- acceptance-test procedure and forms
- list of unresolved engineering assumptions

5. External Technical Review Plan

The goal of external review is criticism, not endorsement. Seek 5-10 contacts across molecular communication, microfluidics, nanosensors, experimental physics, instrumentation, statistics/information theory, quantum measurement, and SETI/technosignatures.

Core reviewer questions:

1. Is there a known conventional physical pathway that invalidates the isolation logic?
2. Are the positive, negative, sham, delayed, replay, clock, and network controls sufficient?
3. Are the statistical thresholds and multiple-testing controls defensible?
4. Is important prior work missing?

5. Does existing literature make the proposal non-novel?
6. Could another lab reproduce Phase 1 from the package?
7. Which assumptions require direct measurement before confirmatory work?
8. What result would justify escalation?

Review disposition categories:

- Critical - must be resolved before release
- Major - normally resolved before release
- Minor - clarity or non-critical implementation improvement
- Deferred - valid but outside current scope
- Rejected with rationale - reviewed but not adopted

6. Publication Plan

Stage 1 - Candidate Release

Use the completed Concept Paper Candidate 1.0 as the publication-review baseline; do not expand it except to answer scientific/reviewer gaps.

Stage 2 - Public Preprint and Research Record

After feedback, publish Concept Paper 1.0 on a DOI-bearing repository such as Zenodo; create an OSF project for protocols, preregistration, data, code, and updates; archive the design package and release hashes.

Stage 3 - arXiv / scholarly distribution

Attempt an appropriate arXiv submission once mature and category/endorsement requirements are addressed.

Stage 4 - Journal or conference submission

Choose the venue after review; the concept paper and later experimental-results paper may require different venues.

7. Build and Experiment Plan

Phase 1A - Known-Physics Commissioning

Prove the challenge generation, microscopic communication, isolation, blinded analysis, telemetry, controls, scoring, and 100-trial commissioning pipeline. No claim about another scale is tested.

Phase 1B - Stronger Isolation / Alternative Microscopic Interface

Only after Phase 1A passes. Change one physical layer at a time while keeping the protocol fixed.

Phase 2 - Search Experiment

Only after commissioning and external review. Remove or block the known information path while retaining blinded challenge/receiver infrastructure. Reproducible challenge-linked information dependence is the target.

Phase 3 - Independent Replication

Any serious candidate must be reproduced by independent laboratories with fresh challenges and independent implementations.

8. Scope-Control Rules

1. Which product does the item belong to?
2. Which defined gap does it close?
3. Is it required before the next project gate?
4. Could it be deferred without weakening the deliverable?
5. Does it introduce a new unvalidated assumption or testing burden?

No feature is added to Phase-1 Design Package 1.0 unless it is necessary to reproduce, validate, isolate, analyze, or safely operate the defined Phase-1 experiment.

9. Change-Control Process

After Candidate 1.0, substantive changes receive an identifier in the form SETI-CR-YYYY-NNN. Each request records the requested change, reason, artifact affected, risk if omitted, new assumptions, verification required, disposition, approver, and date.

10. Immediate Next Actions

1. Complete remaining fabrication/build closure items.
2. Produce Phase-1 Technical Design Package Candidate 1.0.
3. Use the completed Concept Paper Candidate 1.0 as the publication-review baseline; do not expand it except to answer scientific/reviewer gaps.
4. Build an external-review packet with both candidate documents and a one-page reviewer guide.
5. Send to targeted reviewers and log feedback.
6. Resolve Critical and Major findings.
7. Freeze Concept Paper 1.0 and Design Package 1.0.

8. Publish the concept/preprint and repository record.

9. Only then authorize significant Phase-1 hardware procurement and physical build.

11. Definition of Project Success

- a clearly defined and falsifiable search concept
- a reproducible experimental methodology
- a high-quality null result or reproducible anomaly
- public data, protocols, and code sufficient for independent scrutiny
- a clear scientific record separating observation from interpretation

The project should never define success as obtaining a preferred outcome.

12. Final Stop Rule

At that point, we stop expanding documents and start testing the experiment.